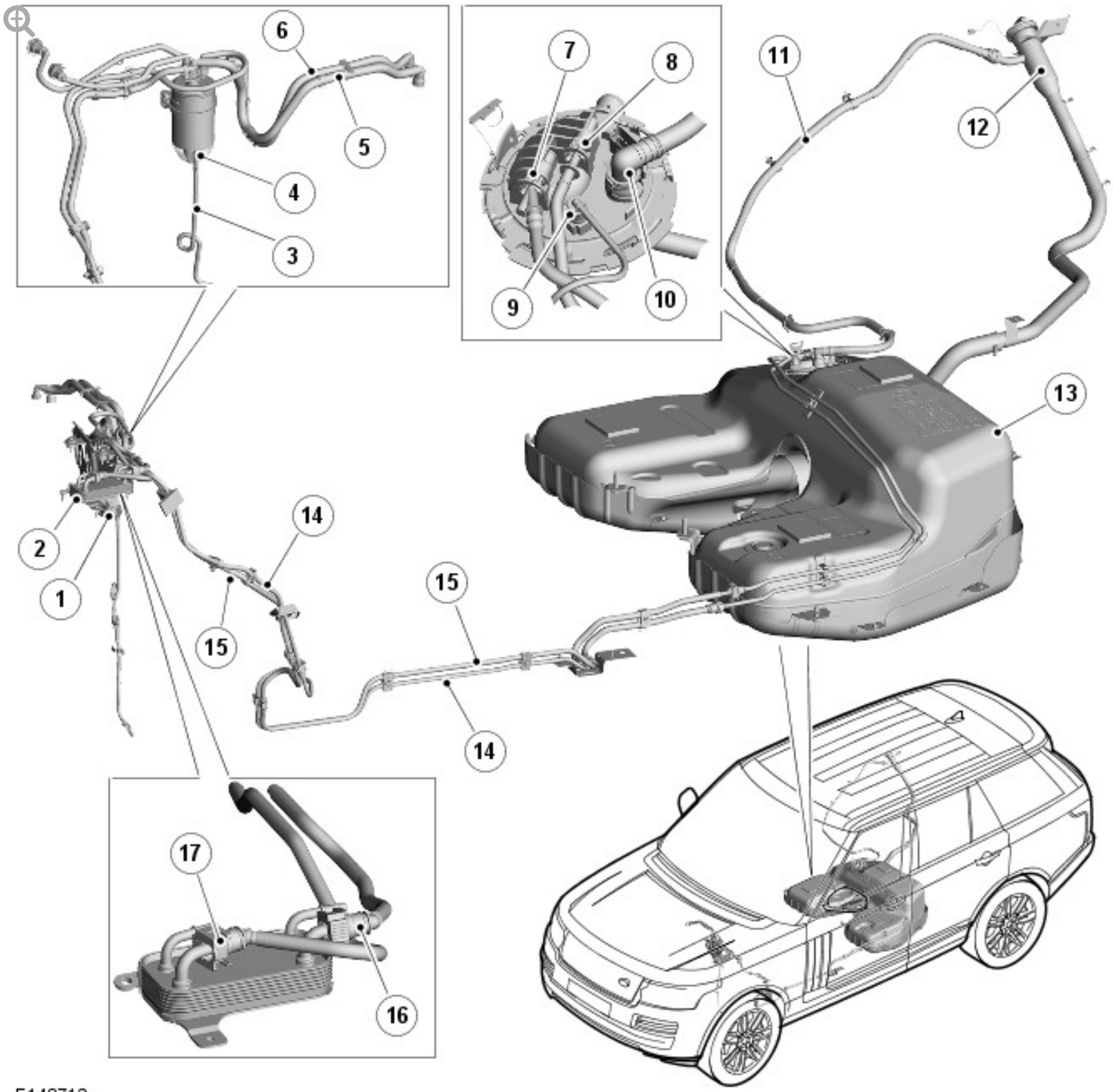
FUEL TANK AND LINES – TDV6 3.0L DIESEL

DESCRIPTION AND OPERATION

COMPONENT LOCATION

TDV6 3.0L DIESEL - FUEL TANK AND LINES



E148712

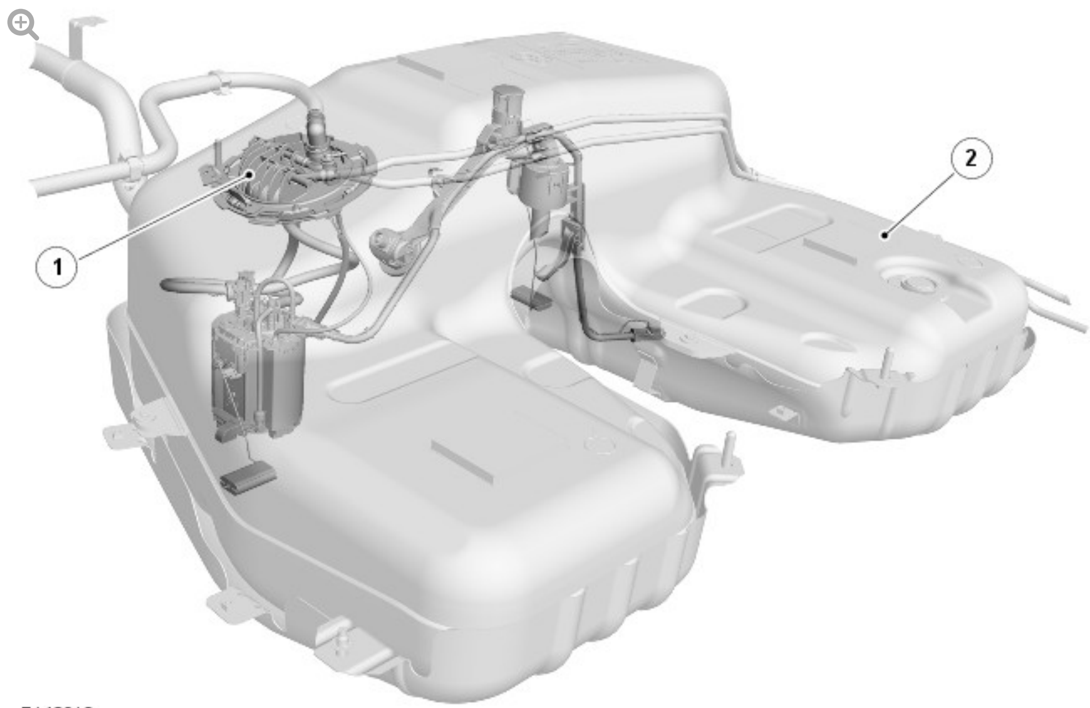
ITEM	DESCRIPTION
1	Fuel Filter
2	Fuel Cooler
3	Water/Fuel Mix Drain Hose
4	Water/Fuel Mix Drain Turn Wheel
5	Fuel Supply to High Pressure Diesel Injector Pump
6	Fuel Return from High Pressure Diesel Injector Pump to Fuel Cooler
7	Fuel Supply to Fuel Filter
8	Fuel Return from Fuel Filter
9	Fuel Fired Booster Heater Supply
10	Fuel Tank Refueling Breather Pipe Connection

11	Fuel Tank Refueling Breather Pipe
12	Fuel Filler Pipe
13	Fuel Tank
14	Fuel Supply Line
15	Fuel Return Line
16	Fuel Cooler to Fuel Filter Pipe
17	High Pressure Diesel Injector Pump to Fuel Cooler Pipe

INTRODUCTION

The TDV6 3.0L diesel engine low pressure fuel system is designed to supply fuel to the high pressure pump, it can supply a uniform level of pressure to the common rails which supply's the fuel injectors.

FUEL TANK



E148913

ITEM	DESCRIPTION

1	Fuel Pump Module Tank Flange
2	Fuel Tank

The fuel tank has a storage capacity of 85 liters.

Inside the fuel tank, you will find the location of the low pressure fuel pump module assembly.

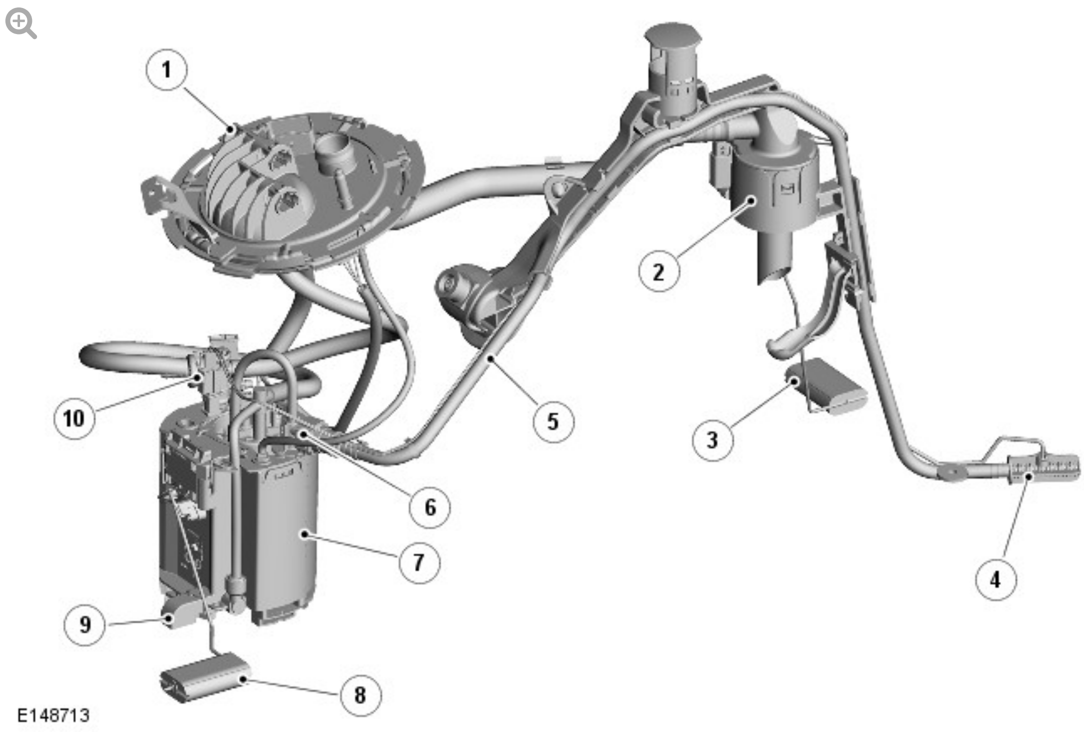
Access to the fuel tank is via the removal of the rear seat base and removal of the low pressure pump module flange located on the right side of the tank.

Within the fuel tank in the upper left side center section, a fuel cut-off elbow is positioned. This is to control the fuel fill volume of the tank. During the Refueling process, air/vapor is displaced (by the fuel entering the tank) and exits the tank via the cut-off elbow and is conveyed by the fuel tank Refueling breather pipe to exit the system. During filling, as the tank reaches its full level, the fuel cut-off elbow is closed by the rising fuel and prevents air/vapor passing through to the Refueling breather pipe. The resulting back pressure causes Refueling to stop automatically.

The fuel cut-off elbow is always open when the fuel tank is below full, providing an unrestricted air/vapor outlet to the Refueling breather pipe.

Fuel level inside the tank is monitored by two level sensors that are connected top the instrument cluster.

FUEL PUMP MODULE



ITEM	DESCRIPTION
1	Fuel Pump Module Tank Flange
2	Fuel Cut-Off Elbow
3	Left Side Fuel Level Sensor Float
4	Suction Port Filter
5	Suction Port Pipe
6	Venturi Pump
7	Swirl Pot
8	Right Side Fuel Level Sensor Float
9	Fuel Pump Coarse Filter
10	Pressure Relief Valve

The fuel pump module is controlled by the ECM (engine control module) via the fuel pump relay located in the RJB (rear junction box).

The electric pump collects fuel from the swirl pot at the base of the pump and passes it from the tank to the fuel supply line to the engine mounted, high pressure pump at a pressure of 0.5 bar.

Fuel is collected from the left side of the tank to the right side of the tank by using a single suction tube that transfers the fuel using a venturi pump which is powered by the fuel feed from the fuel pump.

Should the fuel pump electrical connection need to be disconnected, it is important that the ignition is switched off. If the ignition is on in positions I or II, the instrument cluster will store its last fuel gauge needle position prior to power down. Once power is restored the gauge will display the last stored position regardless of the actual level of fuel in the tank.

This could result in incorrect fuel gauge readings if the fuel tank has been drained and not filled with exactly the same quantity of fuel that was removed.

LOW FUEL STRATEGY

The ECM is programmed with a strategy which shuts the engine off before the fuel tank runs dry. This is to prevent fuel system damage by air being drawn into the high pressure fuel pump.

A misfire is induced with increasing magnitude when the fuel in the tank reaches approximately 2.5 liters of useable fuel remaining to alert the driver to this condition. The engine is shut down when the fuel in the tank reaches 0.00 liters of useable fuel remaining in the tank and 4.0 liters of un-useable fuel remaining.

To reset the fuel strategy after engine shutdown, it is required that a minimum of 4 liters of fuel is added to the fuel tank, when the vehicle is on level ground.

FUEL LEVEL SENSORS

There are two fuel level sensors located inside the fuel tank. One sensor is attached to the swirl pot and monitors fuel level on the right side of the tank. The second sensor is attached to the suction pipe carrier and monitors fuel level on the left side of the tank.

Both level sensors are MAPPS (magnetic passive position sensor) which provide a variable resistance to ground for the output from the fuel gauge.

The electrical output signal is proportional to the amount of fuel in the tank and the position of the float arm. The measured resistance is processed by the instrument cluster to implement an anti-slosh function. This monitors the signal and updates the fuel gauge pointer position at regular intervals, preventing constant pointer movement caused by fuel movement in the tank due to cornering or braking.

Fuel Level Sensors - Resistance/Fuel Gauge Read Out Table

NOTE:

These figures are with the vehicle on level ground. Sensor readings will defer with varying vehicle inclinations.

Active Side of Fuel Tank

SENDER UNIT RESISTANCE, OHMS	NOMINAL GAUGE READING
992	Empty
51	Full

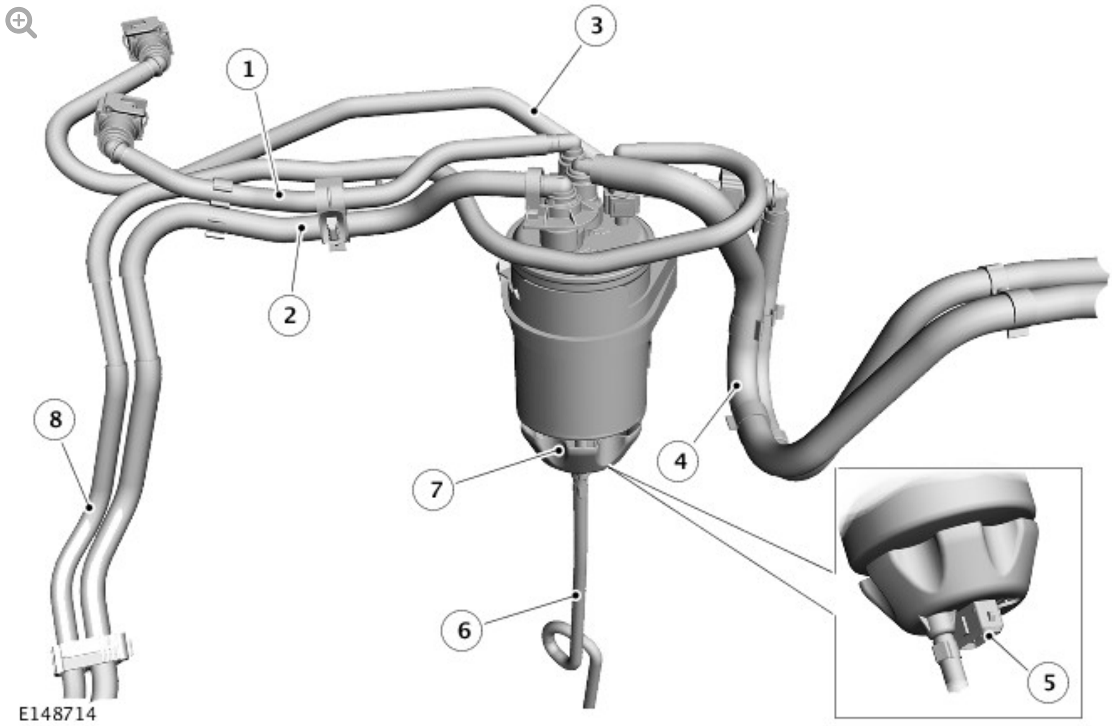
Passive Side of Fuel Tank

SENDER UNIT RESISTANCE, OHMS	NOMINAL GAUGE READING
992	Empty
281	Full

When the fuel level becomes low, a fuel warning lamp in the instrument cluster is illuminated and message is displayed.

The fuel level sender signal is converted into a CAN (controller area network) message by the IC (instrument cluster) as a direct interpretation of the fuel tank contents in liters.

FUEL FILTER



ITEM	DESCRIPTION
1	Fuel Cooler to Fuel Filter Pipe
2	Fuel Supply to Fuel Filter Pipe
3	High Pressure Diesel Injector Pump to Fuel Cooler Pipe
4	Fuel Filter to High Pressure Diesel Injector Pump Pipe
5	Water/Fuel Mix Drain Hose
6	Water in Fuel Sensor

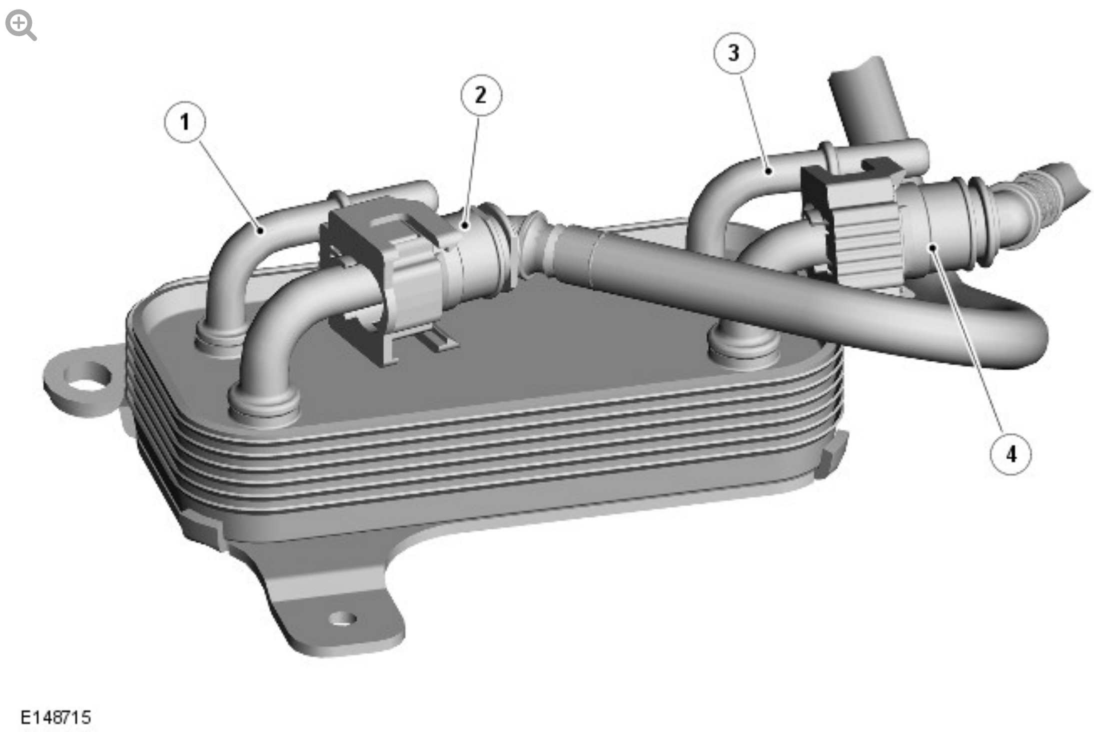
The fuel filter is disposable canister that has four quick release connections.

The fuel filter is fitted with a removable water sensor which is secured at the base of the filter. The sensor can be unscrewed and fitted to a new filter.

The water in fuel sensor housing also incorporates a water drain plastic turn wheel that allows the water/fuel mix to be drained from the filter via the drain hose to the underside of the vehicle.

Water in the fuel filter is sensed by the ECM, the ECM transmits a message on the HS (high speed) CAN powertrain bus to the IC (instrument cluster) which displays a message 'WATER IN FUEL' in the message center.

FUEL COOLER



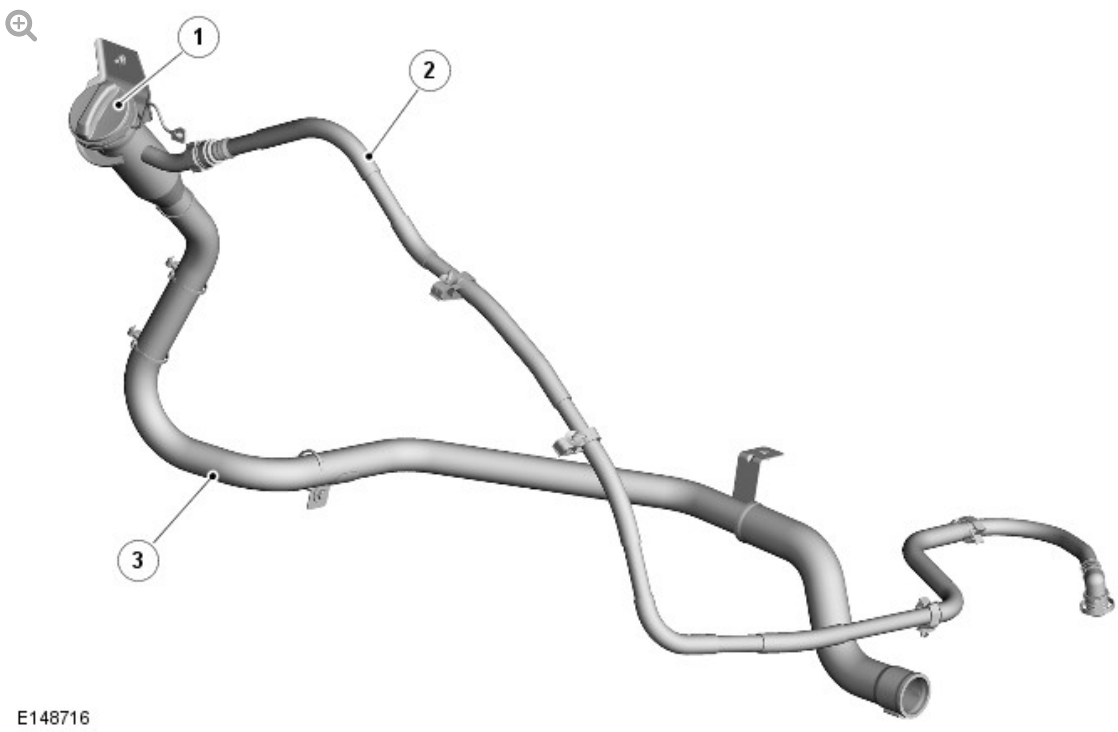
ITEM	DESCRIPTION
1	Coolant Inlet Connection
2	Fuel Inlet Connection
3	Coolant Outlet Connection
4	Fuel Outlet Connection

The fuel cooler is designed to cool fuel returning from the high pressure pump before it is sent back to the fuel tank via the return line. Fuel is cooled by heat transfer through the internal galleries within the coolant assisted fuel cooler.

The fuel cooler is connected to the cooling system. The fuel cooler receives cooled coolant from a dedicated cooler located in front of the engine coolant radiator. This allows for engine coolant to be cooled before it passes through the fuel cooler improving the efficiency of the fuel cooler.

For additional Information, refer to: Engine Cooling (303-03A, Description and Operation)

FUEL FILLER PIPE



ITEM	DESCRIPTION
1	Fuel Cap
2	Fuel Refueling Breather Pipe
3	Fuel Filler Pipe

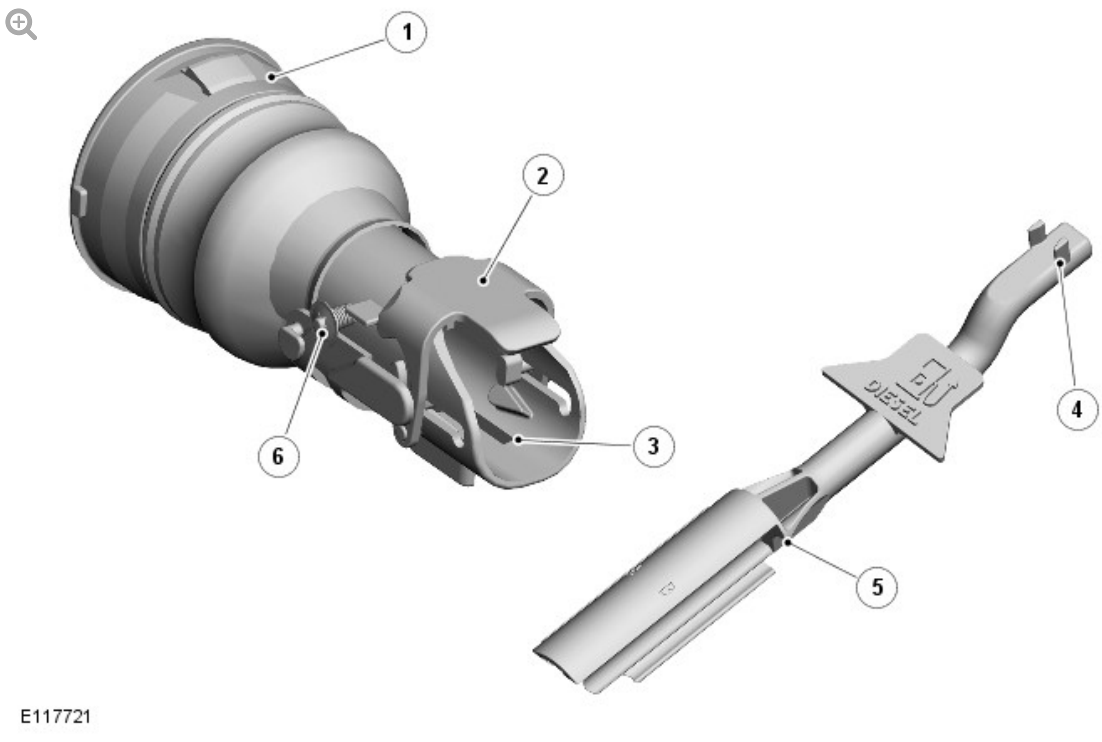
The fuel filler pipe locates into the tank and incorporates a spit back flap in the tank end of the pipe. The flap is spring loaded cover which acts as a one way valve, allowing the tank to be filled but preventing fuel leaving the tank into the filler pipe.

A connection at the top right side of the filler head allows for the connection of the fuel tank Refueling breather pipe.

The filler pipe incorporates a fuel guard system to prevent accidental filling of the tank with petrol.

FUEL GUARD

Passive Fuel Guard System



ITEM	DESCRIPTION
1	Filler neck
2	Flap
3	Reset slots
4	Spigots
5	Reset tool

6	Spring

The passive fuel guard system comprises a mechanically operated flap which is triggered when the smaller diameter filler nozzle tube, used on petrol (gasoline) pumps, is inserted in the filler neck. The shut off flap is actuated and blocks the sensor port on the fuel pump nozzle, causing it to automatically switch off. The shut off flap is locked in this position by a latch mechanism, once it has been activated.

A reset tool is provided and stored within the vehicle. The tool is used to reset the fuel guard device if triggered. The tool is inserted into the filler opening as far as possible with the spigots pointing upper most. Once the two spigots on the tool are located in the slots it can be pulled outwards, releasing a latch and allowing the shut off flap to be opened by its own spring pressure.

The shut off flap is colored yellow so that it is clearly visible within the entrance of the fuel filler pipe when activated, the shut off flap has a 'Handbook' symbol on it.

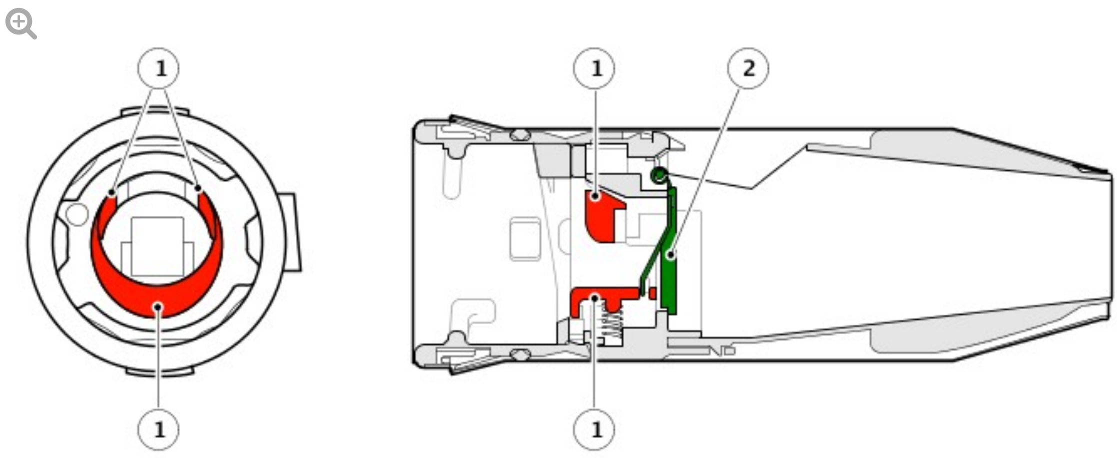
A 25mm diameter diesel fuel pump nozzle will not activate the fuel guard because the nozzle stops against two molded lugs. However, if a 21mm diameter unleaded gasoline pump nozzle is inserted into the housing, its smaller diameter allows it to pass the two molded lugs. The nozzle strikes two pins on the inside of the filler housing which move forward. This movement rotates the shut-off flap which is then held in place when the nozzle is removed by the latch mechanism.

NOTE:

> Russian markets do not use the fuel guard system and are fitted with a conventional filler neck.

Active Fuel Guard System

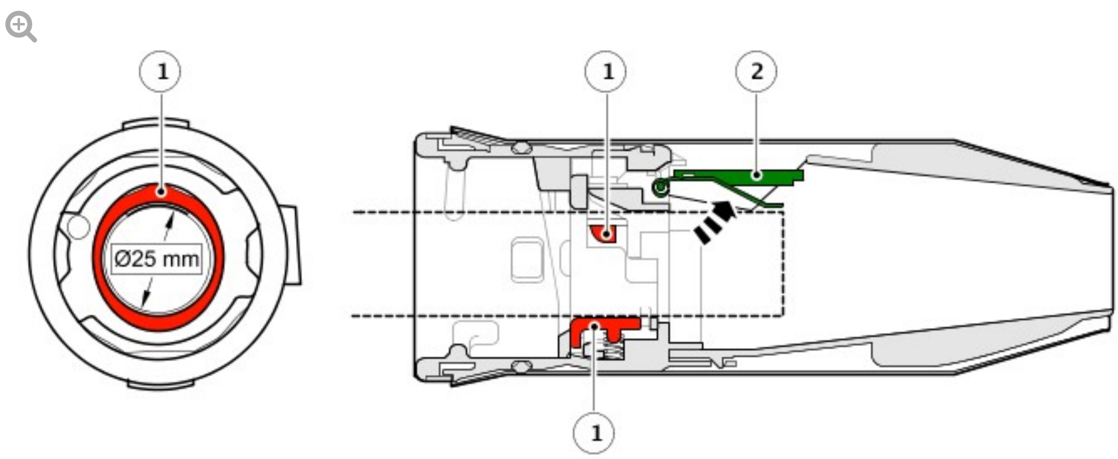
Normal State



E174077

ITEM	DESCRIPTION
1	Sensing tabs in normal state
2	Fuel guard flap closed

Open State

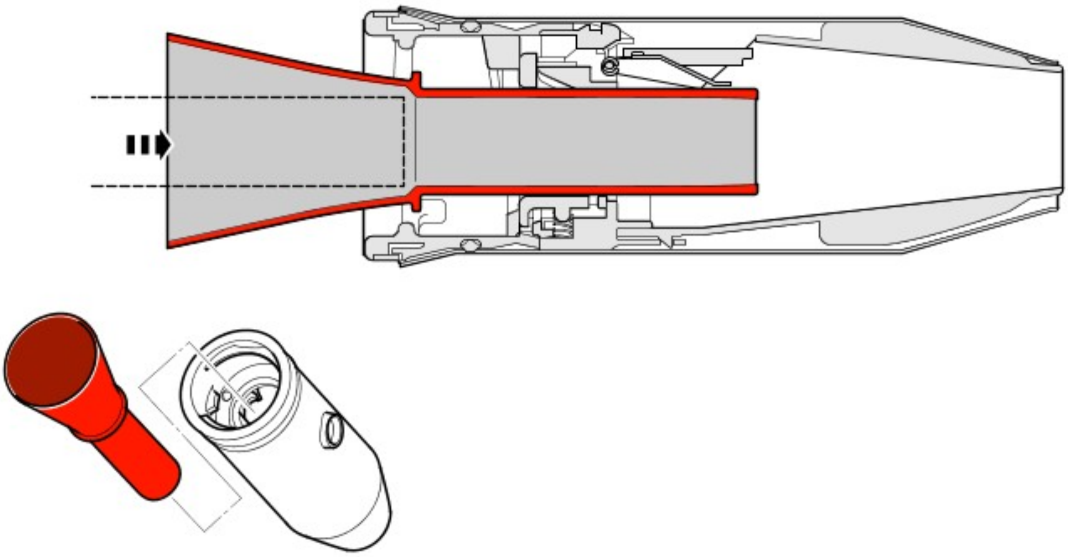


E174076

ITEM	DESCRIPTION
1	Sensing tabs in open state
2	Fuel guard flap open

The fuel guard consists of a mechanically-operated flap and three fuel nozzle sensing tabs. When a 25mm diameter diesel fuel filler nozzle tube is inserted into the fuel filler pipe, the three sensing tabs depress releasing the flap into the open position. If a smaller 21mm diameter filler nozzle tube, used on petrol (gasoline) pumps, is inserted in the fuel filler pipe, the three sensing tabs will not depress and the flap will not release, so this blocks the fuel nozzle from entering the filler pipe.

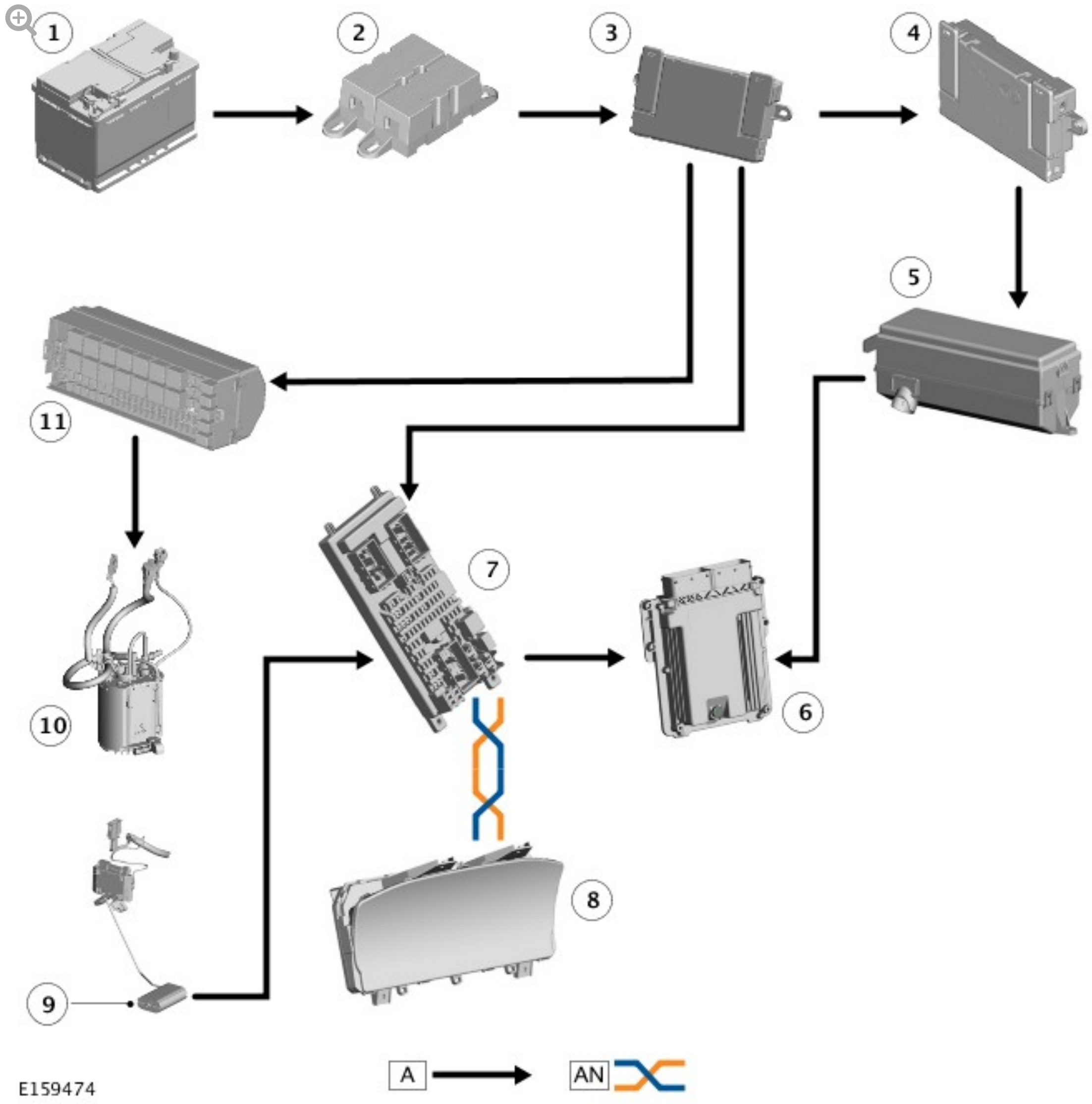
Adaptor Funnel


E174078

If the vehicle travels to a market where differentiation of fuel nozzle size or if the vehicle runs out of fuel and must be refilled at the road side using a reserve can, then an adaptor funnel is provided to allow the misfuelling device to be temporarily overridden.

The adaptor funnel will be located either in the glove compartment or in the luggage compartment alongside the tool kit.

CONTROL DIAGRAM



ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Auxiliary Junction Box (AJB)
5	Engine Junction Box (EJB)
6	Engine Control Module (ECM)
7	Central Junction Box (CJB)
8	Instrument Cluster (IC)
9	Fuel Level Sensor (2 Of)
10	Low Pressure Fuel Pump

11	Rear Junction Box (RJB)